

RAG Setup Guide with Installation Commands

Install Python

Download Python and enable 'Add Python to PATH'.

```
python --version
```

Create Project Folder

Create and enter the project folder.

```
mkdir RAG
cd RAG
```

Create Virtual Environment

Create an isolated Python environment.

```
python -m venv venv
```

Activate Virtual Environment (Windows)

Activate venv.

```
.\venv\Scripts\Activate
```

Upgrade pip (Optional)

Upgrade pip to latest version.

```
python -m pip install --upgrade pip
```

Install Required Libraries

Install all dependencies.

```
pip install sentence-transformers
pip install faiss-cpu
pip install numpy
pip install requests
```

Install Ollama

Download Ollama from ollama.com/download and verify.

```
ollama --version
```

Download Llama 3

Pull the model locally.

```
ollama pull llama3
```

Test Ollama

Verify the model runs.

```
ollama run llama3
```

Run the Application

Execute the RAG application.
`python app.py`

Project Structure

```
RAG/  
├── venv/  
├── app.py  
└── data.txt
```